

THE CELL

THE CELL IS THE BASIC UNIT OF THE BODY. IT IS THE SMALLEST PART CAPABLE OF THE PROCESSES THAT DEFINE LIFE, SUCH AS REPRODUCTION, MOVEMENT, RESPIRATION, DIGESTION, AND EXCRETION – ALTHOUGH NOT ALL CELLS HAVE ALL THESE ABILITIES.

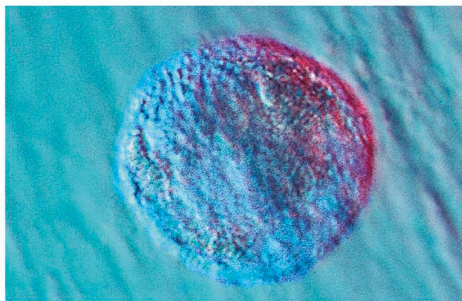
CELL ANATOMY

Most cells are microscopic – a typical cell is 20-30µm in diameter, which means 40 in a row would stretch across a full stop. Very specialized, long, thin cells include neurons (nerve cells) and muscle fibre

cells (myofibres), which may be more than 30cm (12in) long. Most cells have an outer flexible “skin”: the cell, or plasma, membrane. Inside are structures known as organelles, each with a characteristic shape, size, and function. These organelles do not float about at random. The cell is highly organized, with interior compartments linked by sheets and membranes and held in place by a flexible, lattice-like “skeleton” of even tinier tubules and filaments.

EMBRYONIC STEM CELL

Stem cells are unspecialized “beginner” cells that can develop into specialized cells. Stem cells in the embryo can develop into any of the 200-plus types of specialized cells in the body.



CELL TYPES

Cells come in many shapes and sizes, depending on their specialized functions within tissues. Speed of cell division also varies. It is most rapid in epithelial (covering and lining) cells, which are subjected to

physical abrasion and wear and which must continually replace themselves. It is slow or even nonexistent in some cells that are structurally complex, such as nerve cells (neurons).

Epithelial cells

These cells form skin, cover most organs, and line hollow cavities such as the intestinal tract.



Photoreceptor cell

A cone cell is a type of light- and colour-sensitive cell in the retina of the eye.



Red blood cell

The double-dished (biconcave) red cell (erythrocyte) is a bag of oxygen-carrying haemoglobin molecules.



Adipose (fat) cell

The main adipose cells, adipocytes, are bulky and crammed with droplets of fat (lipids), which store energy.



Smooth muscle cell

The large, elongated, spindle-like cells of smooth muscle contract by sliding strands of protein inside.



Nerve cell

Each cell has short branches (dendrites) to receive nerve signals, and a long “wire” (axon) to send them.



Sperm cell

Each sperm has a head that carries the paternal genetic material, and a whip-like tail for propulsion.



Ovum (egg) cell

These giant cells contain the maternal genetic material, and energy resources for the embryo's first cell divisions.

